Weekly Assignment: Sessions 5 & 6

Secure Bedrock Foundation and GPU Model Serving on Amazon EKS

Estimated time 60–90 minutes | Total marks: 20 | Difficulty: Beginner-friendly

Purpose

This assignment checks your understanding of the secure AI workload foundation from Session 5 and the GPU scheduling, model serving and autoscaling flow from Session 6. Use the supplied scenarios and your lab notes. You do not need to build a new AWS environment for this assignment.

What you will practice

- Review S3 privacy, SSE-KMS encryption, VPC endpoint configuration and least-privilege permissions for a Bedrock workload.
- Diagnose simulated AccessDenied errors without weakening security.
- Explain how a GPU-requesting Pod, Karpenter and NVIDIA GPU support work together.
- Explain KServe model serving and KEDA/Prometheus-based replica scaling, including safe cleanup.

Cost and safety rule Do not create new GPU instances or run another load test just to complete this assignment. Do not run terraform apply. Use the supplied evidence scenarios or sanitized outputs from your class lab. Never delete shared EKS/Karpenter/KServe/KEDA/Prometheus resources.

Part A — Session 5: Secure Bedrock foundation (8 marks)

Task A1: Review the security settings (4 marks)

Read starter-files/bedrock-review-scenario.md. Identify four settings that do not meet the stated Lab 5 target. For each, write (1) the observed setting, (2) the risk in one sentence, and (3) the correction. Do not change AWS resources.

Task A2: Diagnose two access failures (4 marks)

Read starter-files/failure-scenarios.md. For each simulated failure, state the most likely cause supported by the error and two safe checks you would perform. Explain why the solution is not to make the bucket public or grant broad permissions such as bedrock:* or s3:* on all resources.

Part B — Session 6: GPU scheduling and serving (10 marks)

Task B1: Explain the workload path (4 marks)

Complete part-b/gpu-provisioning-flow.md using your own words. Put the components in the right order and explain the job of Kubernetes scheduling, Karpenter, the NVIDIA device plugin, KServe, Prometheus and KEDA. Make the distinction clear: Karpenter scales nodes; KEDA scales model Pods/replicas.

Task B2: Interpret a Pending GPU Pod (3 marks)

Use starter-files/gpu-pending-scenario.txt. State whether Pending immediately means the lab is broken, explain what the nvidia.com/gpu: "1" request asks for, and provide two read-only commands from Lab 6 that help investigate if the Pod does not progress.

Task B3: Check autoscaling and clean up safely (3 marks)

Use starter-files/kserve-autoscaling-scenario.txt. Name two checks to make if replicas do not increase during load, explain the expected minimum and maximum replicas, and list the correct cleanup commands. State at least two shared components participants must not delete.

Submission instructions

Copy the submission-template folder and rename it YourName_Sessions5-6_Assignment. Complete all five response files and README.md, then compress the folder into a single ZIP file named YourName_Sessions5-6_Assignment.zip and upload it to the course portal.

Work	What to submit	Marks
A1: Bedrock security review	Four findings, risks and corrections	4
A2: Access-failure diagnosis	Cause and two safe checks for each simulated error	4
B1: GPU provisioning and serving flow	Component sequence and role explanations	4
B2: Pending GPU Pod	Interpretation and two useful read-only checks	3
B3: KServe/KEDA scaling and cleanup	Two autoscaling checks, replica range and cleanup safety	3
Submission quality	Readable files, checklist and safe evidence	2
TOTAL		20

Submission checklist

- ☐ README.md includes my name and a completed-task checklist.
- ☐ All five response files are completed in my own words.
- ☐ Evidence is sanitized; screenshots are optional.
- ☐ No credentials, access keys, account-sensitive logs, Terraform state, raw plan files or other secrets are included.
- ☐ I have not deleted or changed shared trainer infrastructure.

Screenshots are optional. Clear written answers are acceptable. Where you use actual lab output, redact account IDs, names or other details that should not be shared. If you could not access a live environment, use the supplied simulated evidence; do not use personal AWS credentials to work around access restrictions.